

Project Specification: Room Rental Management App

Project Description

This project is a web application designed to help small property owners manage and advertise rental rooms. The idea is based on a real-world need. My parents rent out rooms but currently do not have a dedicated system or website to manage listings and communicate with renters.

The application will allow landlords to create and manage room listings, track availability, and communicate with potential renters. At the same time, renters will be able to browse available rooms, view details, and contact landlords.

Problem it solves:

Small landlords often rely on spreadsheets, messaging apps, or third-party platforms that may be complicated or expensive. This application provides a simple, centralized solution for managing rentals and connecting with tenants.

Target Users

Primary Users:

Small landlords renting out rooms

Secondary Users:

Individuals looking to rent a room such as students, workers, and travelers

Core Features (MVP)

User Accounts:

Users can sign up and log in

Users can manage their profile information

Landlord Features:

Create, edit, and delete room listings

Add room details such as price, location, and description

Mark rooms as available or unavailable

Renter Features:

Browse all available rooms

View detailed information about each room

Contact the landlord through a form or message

Required Data (CRUD Operations):

Users (create, read, update, delete)

Rooms or listings (create, read, update, delete)

Additional / Advanced Features

The application will include at least one advanced feature:

Role-Based Access Control (RBAC):

Users will have roles such as landlord or renter

Different features will be available depending on the role

Optional future enhancements:

Booking or reservation system

Image uploads for room listings

Reviews and ratings

Notifications for inquiries or bookings

Wireframes / Mockups

The application will include the following main views:

1. Home Page
 - Displays a list of available rooms
 - Includes basic search or filtering options
2. Login / Signup Page
 - Allows users to create accounts or log in
3. Room Details Page
 - Displays full information about a room
 - Includes images, description, and contact option
4. Landlord Dashboard
 - Shows the landlord's room listings
 - Provides options to add, edit, or delete listings

Wireframes will be included as sketches or simple mockups.



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Find your perfect room in London.

Welcome Back!

Login to your account

Email

Password

Login

or

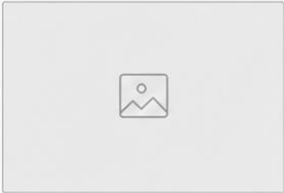
Don't have an account?

Sign Up

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HomeMy BookingsMessages

Back to Listings



Cozy Room in London

£600 / month

London, UK

About this room

Fully furnished room with a comfortable bed, wardrobe, study desk and chair. High speed WiFi included. Close to public transport, shops and restaurants.

Furnished

WiFi Included

Close to Transport

All Bills Included

Contact Landlord

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HomeAbout UsContactLoginSign Up

Search rooms by location, name, etc...

Filter



Cozy Room in London

London, UK

£600 / month

View Details



Modern Studio Room

London, UK

£750 / month

View Details



Shared Flat Room

London, UK

£500 / month

View Details

Trusted Landlords
Verified and reliable

Quality Living
Comfortable and safe

Support
We are here to help

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Landlord

DashboardMy ListingsAdd New RoomBookingsMessagesProfileSettingsLogout

My Listings

Room	Price	Status	Actions
 Cozy Room in London	£600 / month	Available	 
 Modern Studio Room	£750 / month	Available	 
 Shared Flat Room	£500 / month	Rented	 

Quick Overview



3

Total Listings



5

Total Bookings



2

New Messages



£1,850

Monthly Income

Initial Data Models

User:

id
name
email
password
role (landlord or renter)

Room:

id
title
description
price
location
availability (true or false)
landlordId

Booking (Optional):

id
userId
roomId
status (pending, approved, rejected)

API Endpoints

User Endpoints:

POST /users → create a new user account
POST /login → authenticate user
GET /users/:id → retrieve user profile

Room Endpoints:

GET /rooms → retrieve all rooms
GET /rooms/:id → retrieve room details
POST /rooms → create a new room listing
PUT /rooms/:id → update a room listing
DELETE /rooms/:id → delete a room listing

Booking Endpoints (Optional):

POST /bookings → create a booking request
GET /bookings → retrieve booking data

AI Usage

I used AI (ChatGPT) to help brainstorm and organize my project specification. Specifically, I used it to refine my project idea, structure the document, and generate initial data models and API endpoint ideas. I reviewed and edited all generated content to ensure it accurately reflects my project goals and meets the assignment requirements.

Summary

This project provides a simple and practical solution for small landlords to manage rental properties while also giving renters an easy way to find and inquire about available rooms. It focuses on clear functionality, real-world usability, and flexibility for future improvements.